

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite & LIIT **Paper -1(Adv-2020-P1-Model)** **Date: 15-03-2026**

Time: 09.00Am to 12.00Pm **GTA-18** **Max. Marks: 198**

15-03-2026_Sr. XL-500, Sr.S60(N&S), Elite & LIIT_Jee-Adv(2020-P1)_GTA-18_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

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Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCE-2020-P1-Model

Time: 3:00Hour's **IMPORTANT INSTRUCTIONS** **Max Marks: 198**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total mark
Sec – I(Q.N : 1 – 6)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 7 – 12)	**Questions with Multiple Correct Choice +1 partial marks**	4	-2	6	24
Sec – III(Q.N : 13 – 18)	Questions with Numerical Value Answer Type	4	0	6	24
Total				**18**	**66**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 25 – 30)	**Questions with Multiple Correct Choice +1 partial marks**	4	-2	6	24
Sec – III(Q.N : 31 – 36)	Questions with Numerical Value Answer Type	4	0	6	24
Total				**18**	**66**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 42)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 43 – 48)	**Questions with Multiple Correct Choice +1 partial marks**	4	-2	6	24
Sec – III(Q.N : 49 – 54)	Questions with Numerical Value Answer Type	4	0	6	24
Total				**18**	**66**

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PHYSICS **Max Marks: 66**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **SIX** (06) questions.
• Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
• For each question, choose the option corresponding to the correct answer.
• Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +3 If ONLY the correct option is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -1 In all other cases

1. What's the minimum length (non zero) of a space elevator built at the equator, so that it doesn't collapse under gravity ? Consider the elevator to be just a few straight ropes leading to space. The elevator is homogeneous and doesn't contain any extra weight at the end. (Given M = mass of Earth, R = radius of Earth, $\omega =$ angular velocity of rotation of Earth)

A) $h=\sqrt{\frac{2GM}{R\omega^2}+\frac{R^2}{4}}-\frac{R}{2}$ **B)** $h=\sqrt{\frac{2GM}{R\omega^2}+R^2}-\frac{3R}{2}$

C) $h=\sqrt{\frac{GM}{R\omega^2}+R^2}-\frac{3R}{2}$ **D)** $h=\sqrt{\frac{2GM}{R\omega^2}+\frac{R^2}{4}}-\frac{3R}{2}$

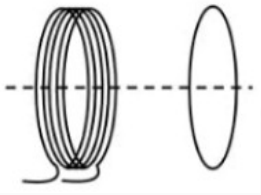
2. Four persons K, L, M and N are initially at the four corners of a square of side a. Each persons now moves with a uniform speed v in such a way that K moves always directly towards L, L directly towards M, M directly towards N and N directly towards K. Find the path-equation followed by the persons. Considering centre of square as origin. (r is the positon of particle from the centre of square).

A) $r=\frac{a}{\sqrt{2}}e^{-\theta}$ **B)** $r=\frac{a}{2}e^{-\theta}$ **C)** $r=\frac{a}{2}e^{-2\theta}$ **D)** $r=\frac{a}{\sqrt{2}}e^{-\frac{\theta}{2}}$

3. An inductance coil consists of several identical turns made of very thin wire. Wound tightly next to each other (see figure). One the axis of the coil, at some distance from it, another identical closed turn is placed, with its axis coinciding with the axis of the coil. The coil is connected to the output of an alternating current source. In this case, the amplitude of the current in the separately placed turn turned out to be K times smaller than the amplitude of the current through the coil. By how many times do the inductances of the coil (measured without the additional turn and together with it) differ ? The resistance of the wire from which the turns are made is negligible. Assume that the inductance of the coil without the additional turn is n times greater than the inductance of a single turn.

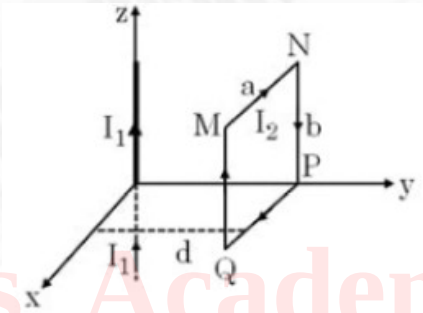
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A) $\frac{1}{1-\frac{1}{nK^2}}$ **B)** $\frac{1}{1-\frac{2}{n}K^2}$ **C)** $\frac{2}{1-\frac{1}{n}K^2}$ **D)** $\frac{1}{2-\frac{1}{n}K^2}$

4. In the figure shown, a very long straight wire carrying current I_1 lies along the z-axis. A rectangular loop MNPQ carrying currying current I_2 is placed such that its plane is parallel to the z-x plane and at a distance d from the z-axis. The side PQ lies in the x-y plane, and the side PN lies in the y-z plane. If F_x *and* F_y are the x-and y-components of the force acting on the loop, then what is the ratio $\frac{F_y}{F_x}$?

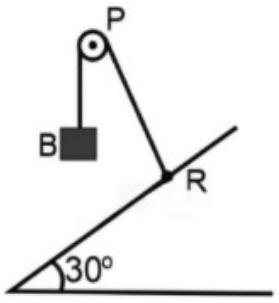


A) $-\frac{a}{d}$ **B)** $-\frac{d}{a}$ **C)** $-\sqrt{\frac{a^2+d^2}{d}}$ **D)** $-\frac{d}{\sqrt{a^2+d^2}}$

5. A small ring R of mass 0.2 kg is threaded onto a fixed rod which is inclined at an angle of 30° to the horizontal. The ring is attached to a block B of mass 0.5 kg by means of a light string which passes over a smooth pulley P. The part RP of the string is perpendicular to the rod, and the part PB of the string is vertical (see diagram). The system is in equilibrium. Assuming the equilibrium is limiting, the coefficient of friction (μ) between R and the rod is_____

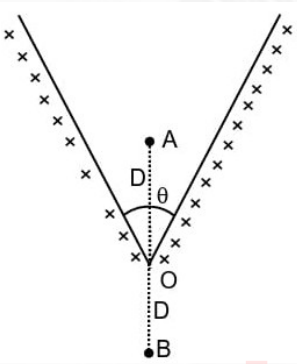
(take $\sqrt{3}=1.75$)

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A) $\frac{4}{13}$ **B)** $\frac{1}{4}$ **C)** $\frac{1}{5}$ **D)** $\frac{1}{10}$

6. In the shown figure an infinitely long V shaped, uniformly charged wire is place. Linear charge density on it is λ. The ratio of the electric field intensity (magnitude) at A and B is: (AB bisects the angle and points A and B are at equal distances from the point O)



A) 2 **B) 1** **C)** $\cos\left(\frac{\theta}{2}\right)$ **D)** $\tan\left(\frac{\theta}{2}\right)$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

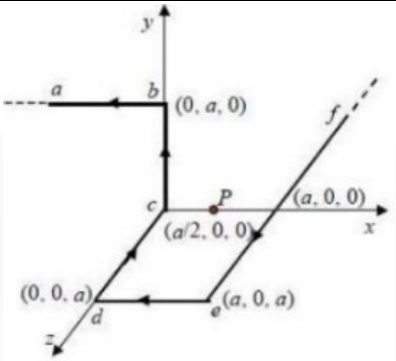
- This section contains **SIX** (06) questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +4 If only (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases

7. A very long wire abcdef carrying current has the shape as shown in the figure. The magnetic field at point P in terms of unit vectors.

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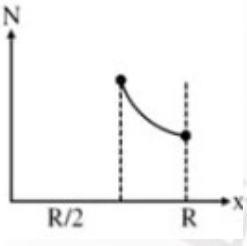
A) Due to wire ab is $\frac{\mu_0 I}{4\pi a}\left[1-\frac{1}{\sqrt{5}}\right]\hat{k}$

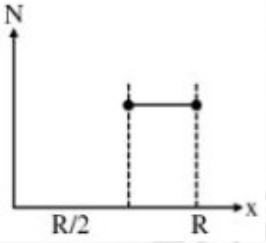
B) Due to wire bc is $\frac{-\mu_0 I}{\pi a\sqrt{5}}\hat{k}$

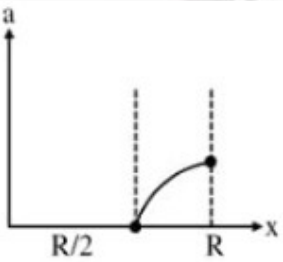
C) Due to wire cd is $\frac{-\mu_0 I}{\pi a\sqrt{5}}\hat{j}$

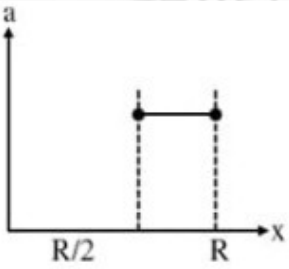
D) Due to wire of ef is $\frac{-\mu_0 I}{2\pi a}\left[1+\frac{2}{\sqrt{5}}\right]\hat{j}$

8. A tunnel is dug along a chord of the earth at a perpendicular distance $\frac{R}{2}$ from the earth's centre. The wall of the tunnel may be assumed to be frictionless. A particle is released from one end of the tunnel. The force N exerted by the particle on the wall and the acceleration a of the particle varies with x (distance of the particle from the centre of earth) according to :

A) 

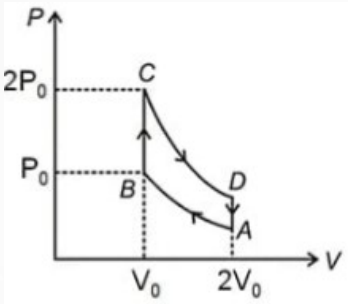
B) 

C) 

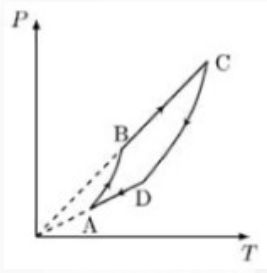
D) 

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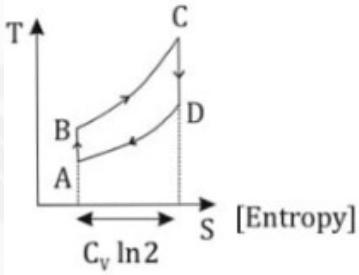
9. One mole of an ideal gas ($c_p / c_v = \gamma$ where symbols have their usual meanings) is subjected an Otto cycle (A-B-C-D) as shown in the following P-V- diagram. Path A-B and C-D are adiabats. The temperature at B is $T_B = T_0$. Choose the correct corresponding graph(s). (symbols have usual meaning)



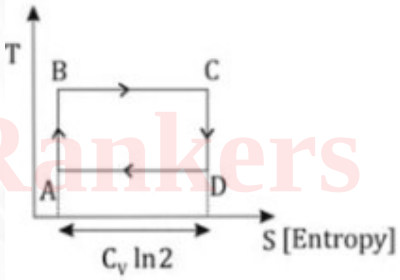
A)



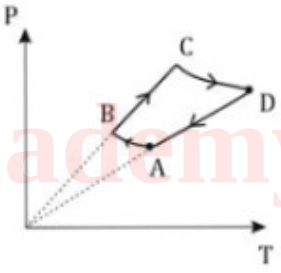
B)



C)



D)



10. Current density in a cylindrical wire is varying with radial distance r as $J = \frac{J_0}{R^2} r(R - r)$, where R is radius of wire. If B is magnetic field due to this wire, then choose the incorrect options.

A) Maximum magnetic field will be at a distance $\frac{8R}{9}$

B) Maximum magnetic field will be at a distance R.

C) Maximum magnetic field will be $\frac{\mu_0 J_0 R}{12}$

D) Maximum magnetic field will be $\frac{64_0 \mu_0 J_0 R}{729}$

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11. The magnetic field associated with an electromagnetic wave propagating in a dielectric medium is given by $\vec{B}=10^{-7}\left(\hat{x}+\hat{y}\right)\sin\left[2\pi\left(5\times10^{14}t-\frac{10^7}{3}z\right)\right]T$. Which of the following option(s) is(are) corret? [Given: The speed of light in vaccum, $c=3\times10^8\,ms^{-1}$]

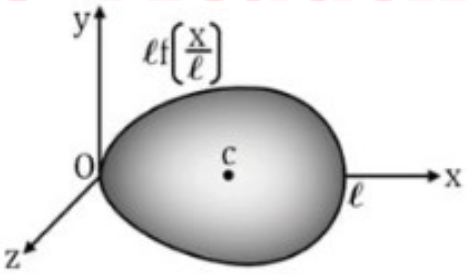
A) $E_x=-15\sin\left[2\pi\left(5\times10^{14}t-\frac{10^7}{3}z\right)\right]v/m$

B) $E_y=-15\sin\left[2\pi\left(5\times10^{14}t-\frac{10^7}{3}z\right)\right]v/m$

C) The wave is polarized in the xy-plane with electric field at an angle of 45° with respect to the positive x-axis

D) The refractive index of the medium is 2.

12. Consider an egg represented by a homogeneous solid of revolution with profile $f(t)=\frac{1}{2}\sqrt{t-t^4}$ on domain $t\in[a=0,b=1]$. To transform this profile into physical units, a length factor l is applied $\left[y=l\,f\left(\frac{x}{l}\right)\right]$. The egg has a density ρ. The egg's centre of mass is at $x=c=\frac{5}{9}l$.



A) The moment of inertia I_{cx} of the egg along the axis x through the centre of mass c is $\frac{1}{288}\pi\rho l^5$

B) The moment of inertia I_{oz} of along the axis z passing through the origin O. is $\frac{115}{4032}\pi\rho l^5$

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C) The moment of inertia I_{cz} of the egg along the axis z passing through the centre of mass c. is $\frac{65}{12096}\pi\rho l^5$.

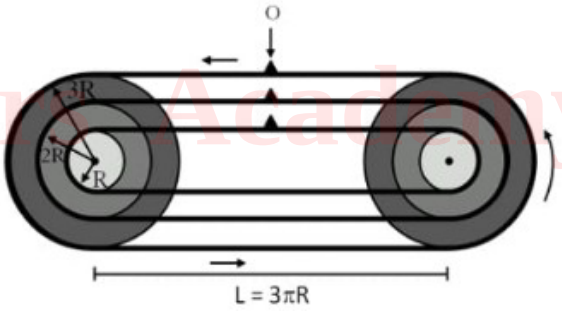
D) The moment of inertia I_{cy} of the egg along the axis y passing through the centre of mass c is $\frac{30}{12096}\pi\rho l^5$.

SECTION – III
(NUMERICAL VALUE TYPE)

- This section contains **SIX** (06) questions. The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. **If** the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme:**

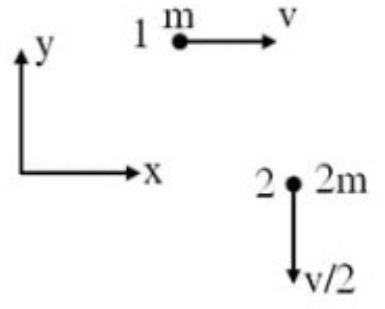
Full Marks: +4 **If** ONLY the correct numerical value is entered;
Zero Marks: 0 In all other cases

13. Two identical shafts are constructed in the form of three concentric cyclinders with radii equal to R, 2R and 3R, and the distance between their centers is $L = 3\pi R$. Both shafts rotate with the same rotation period T_0, and three belts are fastened to the shafts as shown in the figure. On each belt, there is a mark that, at the instant t = 0, is aligned with reference point O. Assuming that the belts rotate without slipping on the shafts, what is the smallest time (in multiple of T_0) after which the three marks are aligned with the reference point O ?



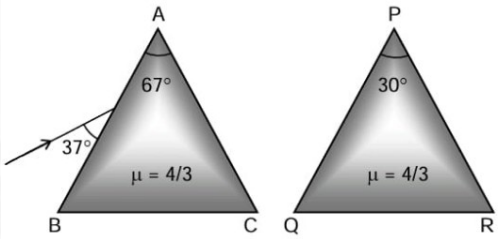
14. Two particles, one of mass m and velocity v, and another of mass 2m and velocity v/2, move perpendicularly on a smooth horizontal surface as shown in figure. At a certain instant, forces of equal magnitude, (and same direction) act on these particles for same time. When these forces stop acting, the first particle acquires a motion perpendicular to its initial direction, with the same speed in +y direction. The magnitude of the velocity acquired by the second particle is $\frac{\alpha v}{2}$, the value of α is:-

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15. A light ray is incident on the edge face AB of a prism ABC as shown in the figure. The second prism is kept in such a manner that emergent ray from prism ABC is falling normally on face PQ of prism PQR. Find the net deviation (in degree) produced in the light ray by the optical system of two prisms.

$(Take\ \sin^{-1}\left(\frac{3}{5}\right)=37^{o}\ \text{and}\ \sin^{-1}\left(\frac{2}{3}\right)=42^{o})$



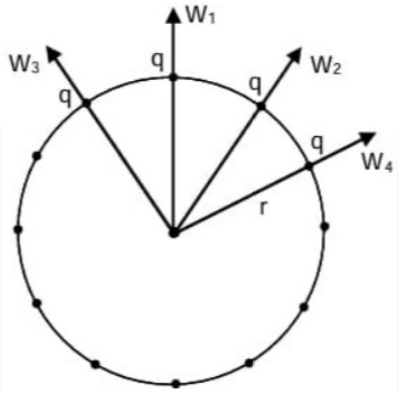
16. A proton is fired from very far away towards a nucleus with charge = 120 e, where e is the electronic charge. It makes a closest approach of 10 fm to the nucleus. The de Broglie wavelength (in units of fm) of the proton at its start is : (assume that nucleus is a rest, take the proton mass

$m_p=(5/3)\times10^{-27}kg; h/e=4.2\times10^{-15}\,J.s./C; \frac{1}{4\pi\varepsilon_0}=9\times10^{9}\,m/F; 1\,fm=10^{-15}m)$

17. N identical charges each of charge 'q' are symmetrically placed on the circumference of circle of radius 'r'. Four adjacent charges are slowly taken to infinity in the sequence as shown in figure. If W_1, W_2, W_3 and W_4 is the respective work done in slowly displacing

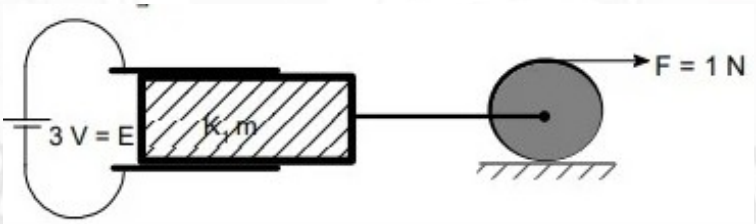
q_1, q_2, q_3 and q_4. If (W_3-W_4) is $\frac{1}{x}\left[\frac{-q^2}{4\pi\varepsilon_0 r\sin\frac{3\pi}{n}}\right]$ *then*

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18. A parallel square plate capacitor of side 'a' is connected to a cell of emf 3 vol. A dielectric of mass m = 0.1 kg and dielectric constant 'K' initially completely fills the space between plate of parallel plate capacitor. The dielectric is connected to a solid cylinder of mass m = 0.1 kg through the string. Cylinder is pulled by a tangential force F =1 N and acting on top of cylinder. The surface between cylinder and horizontal surface is enough rough to prevent slipping. The distance between plate is d. The graph between the current through the cell v.s time 't' is a straight line till dielectric leaves the capacitor. Find the slope of straight line.

$$\left[Given\ \frac{\in_0 a(K-1)}{d}=\frac{1}{9}\right]$$



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CHEMISTRY **Max Marks: 66**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **SIX** (06) questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
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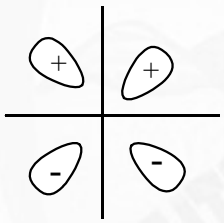
Full Marks: +3 If ONLY the correct option is chosen;
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Negative Marks: -1 In all other cases

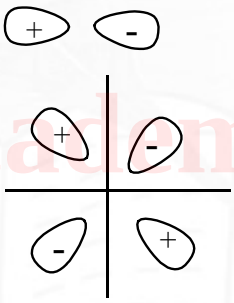
19. $E^0_{A^{2+}/A^+} = 0.1V$; $E^0_{A^{3+}/A^{2+}} = 0.064V$

A fresh $A(NO_3)_2$ aqueous solution is prepared. After some time an equilibrium with A^{2+}, A^{3+} *and* A^+ is reached. What is the mole fraction of A^{2+} at the equilibrium? $(\log_2 = 0.3)$

A) 0.25 **B)** 0.2 **C)** 0.66 **D)** 0.5

20. O_2 molecule has an unpaired electron in the orbital shown as :

A) 

B) 

C) 

D)

21. $A(NO_3)_{2\,(aq)} \xrightarrow{aqKI} X\downarrow + B_{(aq)} \xrightarrow{\Delta} X_{(aq)} + B_{(aq)} \xrightarrow{cooling} X\downarrow + B_{(aq)}$

$X_{(solid)} \xrightarrow{Conc.H_2SO_4} Y\uparrow \xrightarrow{cooling} Y_{(solid)}$

Which statement is correct ?

A) 'A' may be 'Cu' and 'Y' can reduce hypo.

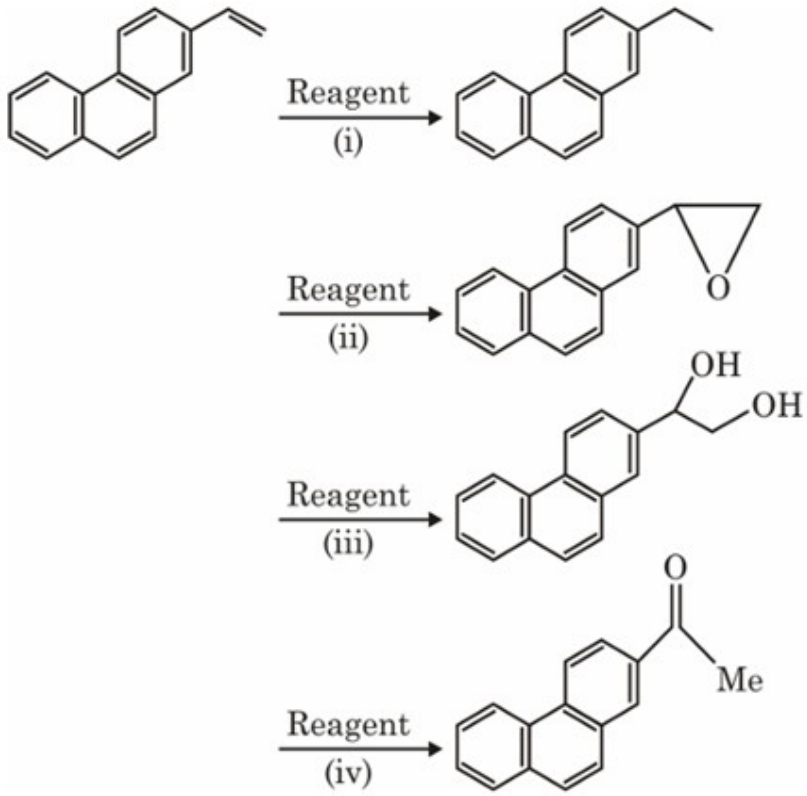
B) 'A' may be Ba and 'Y' is soluble

C) 'A' may be 'Pb' and 'Y' can oxidize hypo

D) If 'A' is lead 'X' is black

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22.



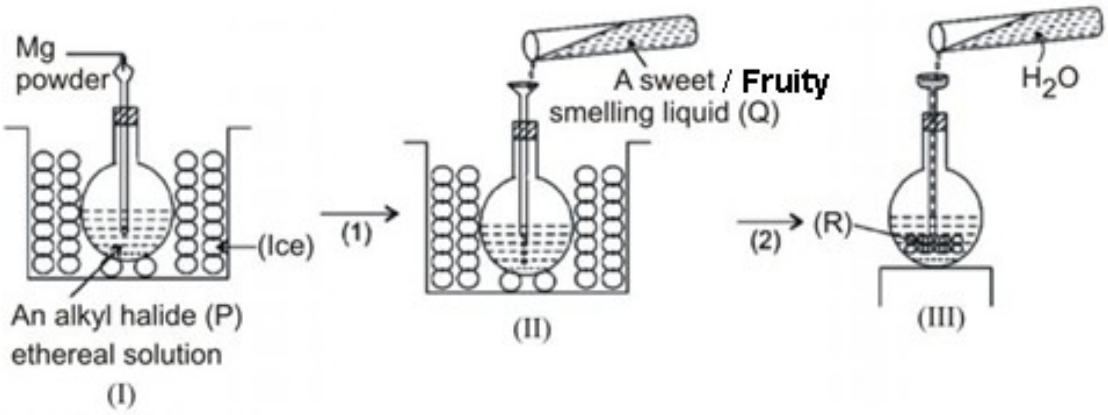
Which is incorrect option for above reactions from following set of reagents (i), (ii), (iii) and (iv) ?

	(i)	(ii)	(iii)	(iv)
w	H_2/Pt	MCPBA	Baeyers reagent	$O_2 + PdCl_2 + CuCl_2 + H_2O$
x	H_2/Pd + $BaSO_4$ + quinoline	HCO_3H	Hot alk. $KMnO_4$	dil. $H_2SO_4 + Hg^{2+}$
y	$Sia_2BH + CH_3COOH$	$PhCO_3H$	OsO_4 / $NaHSO_3$ / H_2O	$O_2 + PdCl_2 + CuCl_2 + H_2O$
z	BH_3 + THF + CH_3COOH	MCPBA	Baeyers reagent	$O_2 + PdCl_2 + CuCl_2 + H_2O$

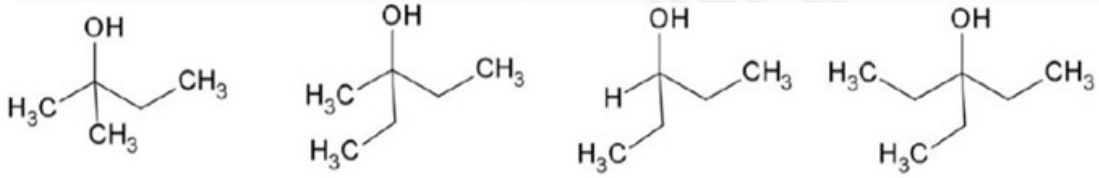
A) w **B)** x **C)** y **D)** z

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23. Consider the following scheme :



If P is ethyl chloride, R can be

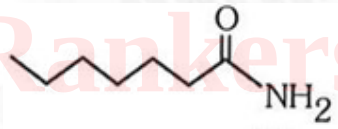


A) I, II, III, IV **B)** II, III, IV only **C)** IV only **D)** II, IV only

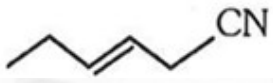
24. The following four compounds are subjected to different reactions

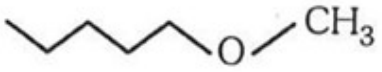
Choose the correct statements for these reactions

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I) 

II) 

III) 

IV) 

A) Compound I on reaction with bromine and NaOH gives the same compound as that obtained by complete reduction of III.

B) All compounds given I, II, III, IV undergoes reduction with $H_2 + Ni$

C) Compound III on reaction with $LiAlH_4$ gives a compound which reacts with bromine $+CCl_4$ under suitable condition gives meso compound.

D) The compound obtained by the reaction of $LiAlH_4$ with II is same as that obtained on heating IV with dilute acid.

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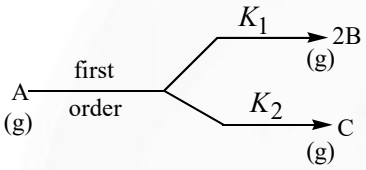

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

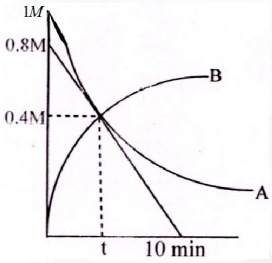
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Full Marks: +4 If only (all) the correct option(s) is(are) chosen;
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Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases

25.



The above reaction gives the following graphical data



Which statements are correct ?

A) The half life of the reaction is approximately 3.5 min.

B) At any stage of the reaction [B] / [C] is equal to one.

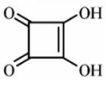
C) The ratio K_1 *to* K_2 is 2

D) At time 't' minutes [c] = 0.2 M

26. Which of the following is/are correct?

A)  The given Compound can undergo Friedel Crafts acylation

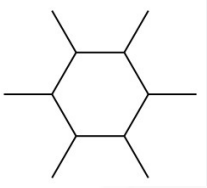
B) Ethyl alcohol reaction with benzene diazonium chloride forms benzene under suitable conditions

C)  Squaric acid can give positive sodium bicarbonate test.

D) During vulcanization of rubber, sulphur bridges (or) cross links b/w polymer chains are introduced either at their reactive allylic positions (or) at the sites of double bonds.

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27. i) Total no. of geometrical isomers possible for 1, 2, 4-tribromocyclohexane are possible is 'x'.

ii)  total number of geometrical isomers possible for given compound is 'y'

values of 'x' and 'y'.

A) x = 4 **B)** y = 8 **C)** x = 3 **D)** y = 9

28. Which statement(s) is (are) correct?

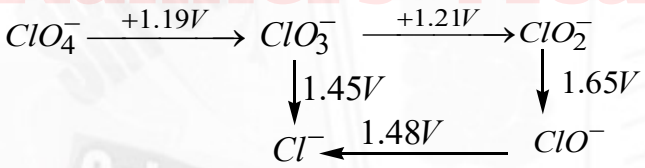
A) In a "mer Ma_3b_3" molecule if any one ligand is replaced by 'c', two isomeric products may be formed.

B) In a "fac Ma_3b_3" molecule if any ligand is replaced by 'c', two isomeric products are formed.

C) $\left[Zn(gly)_2\right]$ has two geometrical isomers

D) During Mond's process to purify Nickel the bond length in the approaching ligand increases in the gas phase complex.

29. Standard reduction potentials are shown in the following sequence.



Which statements are correct in standard conditions?

A) Disproportion of ClO_3^- into ClO_4^- *and* Cl^- is spontaneous

B) ClO^- is a better oxidant than ClO_3^- to form Cl^-

C) ClO_2^- disproportionate into ClO_3^- and OCl^-

D) ClO_2^- / Cl^- half cell acts as anode to ClO^- / Cl^- to produce electric current

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30. In Zinc sulphide crystals, sulphide ions take up cubic close packing structure and zinc ions occupy alternate tetrahedral voids. (edge length of unit cell = a)

Which statement(s) is (are) correct?

A) Nearest 'cation-cation' distance is close to "$a/\sqrt{2}$"

B) Each 'Zn^{2+}' ion has twelve equidistant nearest Zn^{2+} ions.

C) Nearest "anion-anion" distance and nearest cation-cation distance are in 1:1 ratio

D) One mole 'ZnS' contains one mole of unoccupied 'tetrahedral' voids.

SECTION – III
(NUMERICAL VALUE TYPE)

- This section contains **SIX** (06) questions. The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. **If** the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +4 **If** ONLY the correct numerical value is entered;
Zero Marks: 0 In all other cases

31. A metal surface is irradiated by photons of 4.13 eV energy and stopping potential is exactly provided by the cell:

$$Pt:\ \underset{(1atm)}{H_2}\ /\ \underset{\substack{(0.1M)\\ \alpha=1}}{HX}\ //\ \underset{\substack{(0.1M)\\ \alpha=1}}{HX}\ /\ \underset{(1atm)}{X_2}:Pt$$

If energy of the incident radiation is increased to 4.25eV, to provide exact stopping potential, the solution at both compartments need to be diluted by 'n' times each. What is 'n'?

32. '0.25M' HF solution ($k_a = 10^{-2}$) is mixed with equal volume of '0.5 M' KF solution. What is the ratio of degree of dissociation of HF after and before common ion effect? (Assume dissociation of water is totally suppressed)

33. How many grams of $H_2(g)$ should be consumed in a hydrogen fuel cell such that 80% of the maximum derivable work is utilized to raise the temperature of 9.65 moles of a mono atomic ideal gas adiabatically by 50^oC through compression ?

$O_2(g)+4H^+ +4e^- \rightarrow 2H_2O(l)\ E^o = 1.25V$

$2H^+(aq)+2e^- \rightarrow H_2(g)\ E^o = 0.00V$

$F = 96500\,Cmol^{-1},\quad R = 8JK^{-1}mol^{-1}$

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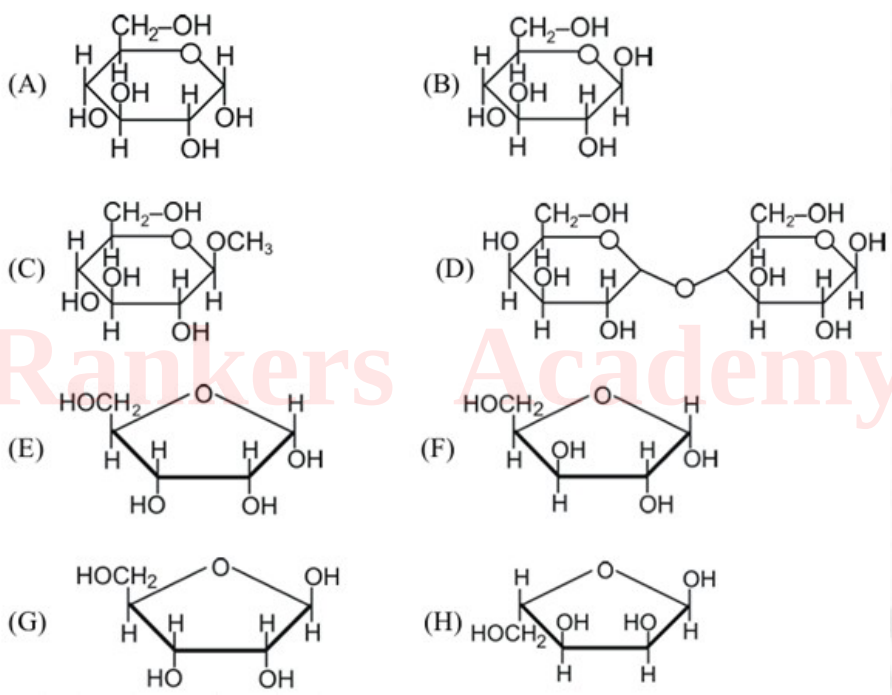

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34. 200 m*l* of a freshly prepared saturated solution of M_2SO_4 shows 1st order radio active disintegration with decay rate of $10^{10}\,dps$ due to "M^{2+}" ions. The K_{SP} of the salt is $A\times10^{-14}$ what is A?

Decay constant of 'M' is $1.66\times10^{-9}\,\text{sec}^{-1}$

35. 0.1 mole of $Hg(NO_3)_2$ and $0.5 moles$ of KCN are mixed in one 'Kg' of water. What is the boiling point (^oC) of the solution at '1atm' pressure. (Ebulioscopic constant of water $=0.5\,k\,kgmol^{-1}$) Note : No complex is formed.

36. From given compounds



From given structures

Total number of reducing sugar = x

Total number of sugars in which molecular weight increases by 210, whenever it undergo acetylation reaction with excess of acetic anhydride = y

Then find the value of x - y = ?

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MATHEMATICS **Max Marks: 66**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **SIX** (06) questions.
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37. $I_1 = \int_0^{\frac{\pi}{2}} \frac{\sqrt{2}\cos\left(x-\frac{\pi}{4}\right)dx}{1+\pi^{\sin\left(x-\frac{\pi}{4}\right)}}$,

$I_2 = \int_0^{\frac{\pi}{2}} \frac{\cos\left(x-\frac{\pi}{4}\right)dx}{1+\left(\frac{\pi}{2}\right)^{\sqrt{2}\sin\left(x-\frac{\pi}{4}\right)}}$

Then the correct relation is

A) $I_1 = I_2$ **B)** $I_1 = I_2\sqrt{2}$ **C)** $I_1\sqrt{2} = I_2$ **D)** $2I_1 = I_2$

38. A bag contains 3 maths books and 2 physics books. A book is drawn at random, if it is of maths, 2 more books of maths together with this book put back in the bag and if it is of physics it is not replaced in the bag. This experiment is repeated 3 times. If third draw gives a maths book, what is the probability that first two drawn books were of physics?

A) $\frac{84}{347}$ **B)** $\frac{21}{347}$ **C)** $\frac{42}{347}$ **D)** $\frac{168}{347}$

39. Let x, y and z be real numbers satisfying the system of equations

$\log_2(xyz-3+\log_5 x)=5,$

$\log_3(xyz-3+\log_5 y)=4,$

$\log_4(xyz-3+\log_5 z)=4,$

Value of $|\log_5 x|+|\log_5 y|+|\log_5 z|$ *is*

A) 225 **B)** 125 **C)** 365 **D)** 265

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40. Let a,b,c be complex numbers such that $|a|=|a-b|=3\sqrt{3}, |a+b+c|=21$ and $a^2+b^2+c^2=ab+bc+ca$, then $|b^2|+|c^2|$ equals

A) 132 **B)** 140 **C)** 147 **D)** 150

41. The equation of the line passing through the point (1, 2, -4)and perpendicular to the two lines $\frac{x-8}{3}=\frac{y+19}{-16}=\frac{z-10}{7}$ *and* $\frac{x-15}{3}=\frac{y-29}{8}=\frac{z-5}{-5}$, will be

A) $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z+4}{6}$ **B)** $\frac{x-1}{-2}=\frac{y-2}{3}=\frac{z+4}{8}$

C) $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z+4}{5}$ **D)** None of these

42. The points D, E, F divide BC, CA and AB of the triangle ABC in the ratio 1:4;3:2 and 3:7 respectively and the point K divides AB in the ratio 1 : 3, then $|\overrightarrow{AD}+\overrightarrow{BE}+\overrightarrow{CF}|:|\overrightarrow{CK}|$ is equal to :

A) 1 : 1 **B)** 2 : 5 **C)** 5 : 2 **D)** None of these

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

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Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
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Negative Marks: -2 In all other cases

43. The first of the two samples has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{13.44}$

A) Mean of second group is 16

B) Mean of second group is 18

C) Standard deviation of second group is 4

D) Standard deviation of second group is 5

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44. Five circles $S_i = 0,\ i = 1,2,3,4,5$ have centers respectively at $C_1(0,0), C_2(2,2), C_3(6,10), C_4(1,-2), C_5(4,-3)$ and the line $y = mx + c$ is common tangent to all five circles such that C_1, C_2, C_3 lie on one side and C_4, C_5 lie on other side of the tangent. It is given that sum of radii of $S_1 = 0$ *and* $S_2 = 0$ is equal to radius of $S_4 = 0$ and also sum of radii of $S_2 = 0, S_3 = 0$ is equal to the radius of $S_5 = 0$ then

A) m = c **B)** m + c = 0 **C)** $m^2 + c^2 = 2$ **D)** $|m| = |c|$

45. Consider a parabola $P: y^2 = 4x$ and a line $L: y = 2x - 4$. L meet P at A and B, with AB as base of an isosceles triangle ABC is drawn whose vertex C lies on x-axis. Which of the following statement(s) is(are) true ?

A) The coordinates of C are $\left(\frac{5}{2}, 0\right)$

B) The coordinates of C are $\left(\frac{9}{2}, 0\right)$

C) Point of intersection of the tangents at A and B is (-3,2)

D) Point of intersection of the tangents at A and B is (-2,1)

46. Two different chords C_1 *and* C_2 of a hyperbola $\frac{x^2}{4} - y^2 = 1$ are drawn from a point P(-2, 0). A line $L: y - mx - c = 0,$ which is tangent to parabola $y^2 = 4x$ bisects C_1 *and* C_2.. which of the following CANNOT be the value of m ?

A) 0 **B)** $\frac{2}{7}$ **C)** $\frac{2}{\sqrt{3}}$ **D)** $-\frac{2}{\sqrt{3}}$

47. Let $f: \mathbb{R} \to \mathbb{R}$ be a function such that $f(x) = [x]^2 + [x+1] - 3,$ where [x] = the greatest integer $\le x$. Then

A) f(x) is a many-one and into function

B) f(x) = 0 for infinite number of values of x

C) f(x) = 0 for only two real values

D) None of these

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48. Let $f:R\to R$ be any twice differentiable function such that its second derivative is continuous and $f'(x)\neq 0$ for all $x\neq 0$ and $\lim\limits_{x\to 0}\dfrac{f(x)}{x^2}=\pi$. then

A) $f(x)>f(0)\forall x\neq 0$ **B)** $f(x)<f(0)\forall x\neq 0$

C) $f(x)$ has local minima at x = 0 **D)** $f''(0)=2\pi$

SECTION – III
(NUMERICAL VALUE TYPE)

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49. If the number of solutions in $[0,\ 2\pi]$ satisfying $\tan^4 x+\cot^4 x+1=3\sin 2x,$ is λ and the number of solutions of the equation

$$1+\cos x+\cos\left(x+\frac{x}{2}+\frac{x}{4}+\frac{x}{8}+....\infty\right)+\cos\left(2x+\frac{2x}{3}+\frac{2x}{9}+\frac{2x}{27}.....\infty\right)=3+\frac{3}{4}+\frac{3}{16}+\frac{3}{64}.....\infty$$ in $[0,\ 4\pi]$ is 't' then $(\lambda+t)$ equals

50. The solution set of $|x-1|^{\log_2(4-x)}<|x-1|^{\log_2(1+x)}$ is (0, a) $\cup$ (b, c) then (abc) has the value equal to:

51. Let a and b be the coefficient of x^2 $in\left(1+x+2x^2+3x^3\right)^4$ $and\left(1+x+2x^2+3x^3+4x^4\right)^2$ respectively. Value of (a-b) equals.

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52. Let A, B, C be three sets of functions:

$$A=\left\{[x],\sqrt{1-x^2},\frac{1}{x-1},\frac{2}{1-\{x\}}\right\},\quad B=\left\{\tan x,\sin x+\cos x,\frac{3}{2+\cos x}\right\},$$

$$C=\left\{x^{\frac{1}{3}},\tan^{-1}x+\operatorname{sgn}\left(1+[x]+[-x]\right)\right\}$$

(where, [.] is greatest integer functions & {.} is fractional part function and sgn(x) is signum function) A cubic, fair die, with faces numbered 1,2,3,4,5,6, is thrown once. If it turns up a composite number, a function is selected from the set A; if it shows up a prime number, a function is selected from the set B, otherwise a function is selected from the set C, If the selected function is found to be derivable in its domain and the probability that it was selected from set A, is $\frac{p}{q}$, where $p,q\in N$, then the least value of $(q-7p)$ is _____

53. Let $f(x)=ax^2+bx+c;\ a,b,c\in N$ and $\int_0^1 f(x)dx=\frac{11}{6}$, then the number of ordered pairs (x, y) of equation $f([x])f([y])=2[x][y]$ is _____ (where [.] denotes the G.I.F

54. The number of subsets 'R' of the set P = {1, 2, 3,…., 9} which satisfy the property "There exist integers a < b < c with $a\in R,b\notin R,c\in R$" is

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